

Abiotic and Biotic Influences on Gorgonian Distribution in the Southern California Bight

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An Honors Thesis Submitted to the Biology Department at Occidental
College, Los Angeles, California, USA on 04/15/2025

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I. Abstract

Temperate reef ecosystems are biodiversity hotspots that provide essential ecological functions. Gorgonians, sessile colonial octocorals, enhance three-dimensional reef complexity, yet their distribution and density in the Southern California Bight (SCB) remain poorly understood. This study examines the environmental and biological drivers of gorgonian populations across natural and artificial reefs in the SCB. Key factors analyzed include giant kelp density, crustose coralline algae (CCA) interactions, substrate heterogeneity, relief heterogeneity, sea surface temperature (SST), and chlorophyll-a concentration. Data were collected by the Vantuna Research Group (VRG) and the Partnership for Interdisciplinary Studies of Coastal Oceans (PISCO) from 126 reef sites using the Co-operative Research and Assessment of Nearshore Ecosystems (CRANE) protocol, incorporating depth-stratified sampling and SCUBA-based surveys. A linear mixed-effects model identified substrate heterogeneity ($\beta = 12.51$, $p < 0.001$) and mean reef relief ($\beta = 9.07$, $p < 0.001$) as significant predictors of gorgonian density, emphasizing the role of structural complexity. Gorgonian density was higher on artificial reefs than natural reefs ($\beta = -42.50$, $p = 0.05$), while giant kelp density and SST showed no significant effects. Competitive interactions with CCA were evident ($\beta = -0.06$, $p = 0.015$). These findings provide a framework for predicting gorgonian distribution in relation to reef structure and environmental conditions. Understanding gorgonian habitat preferences is essential for managing their populations and maintaining reef ecosystem balance, particularly as environmental conditions continue to change. Further research is needed to assess long-term gorgonian-kelp dynamics and potential shifts in reef community composition.

II. Background

Temperate reef ecosystems are biodiversity hotspots that provide essential ecological functions, including nutrient cycling, carbon sequestration, and habitat for a diverse array of

marine species (Bulleri & Chapman, 2010; Smale et al., 2019). These ecosystems are structured by a combination of abiotic and biotic factors, including substrate composition, water temperature, and species interactions (Filbee-Dexter & Scheibling, 2014). Among the key habitat-forming taxa in these systems, gorgonians (Order *Alcyonacea*) play a crucial role in enhancing three-dimensional habitat complexity, providing shelter and attachment surfaces for various reef-associated organisms, including juvenile fishes and invertebrates (Tsounis et al., 2012). Despite their ecological importance, the factors influencing gorgonian distribution and density in the Southern California Bight (SCB) remain poorly understood.

The SCB is a dynamic coastal system that encompasses a variety of reef types, including natural rocky reefs and artificial reef structures, both of which serve as important habitats for benthic communities (Pondella et al., 2018). Environmental gradients such as sea surface temperature (SST), substrate variability, and productivity (measured via chlorophyll-a concentration) shape the distribution of reef-associated taxa, influencing their settlement, growth, and survival (Harvell et al., 1996; Edmunds et al., 2020). In particular, substrate heterogeneity and reef relief have been shown to enhance habitat suitability for sessile invertebrates, including gorgonians, by increasing the availability of stable attachment sites and reducing sedimentation stress (Linares et al., 2008). However, how these factors interact with biological variables, such as competition with macroalgae and exposure to temperature stress, remains an open question in temperate reef ecology.

a. Gorgonians in the SCB: Ecological Context

Gorgonians are sessile colonial octocorals that rely on stable substrates for attachment and growth. They are particularly sensitive to environmental variability, with their distribution

being influenced by hydrodynamics, substrate availability, and competitive interactions with other benthic taxa (Lasker et al., 2020). Natural and artificial reefs in the SCB differ in their structural complexity, which may influence gorgonian population dynamics. Artificial reefs, often designed to enhance fisheries and restore habitat, tend to exhibit higher relief and more heterogeneous substrates compared to natural reefs, potentially providing more suitable settlement sites for gorgonian larvae (Perkol-Finkel et al., 2008; Fowler & Booth, 2013).

Understanding whether reef type plays a significant role in structuring gorgonian populations is critical for evaluating the ecological function of artificial habitats in temperate reef ecosystems.

Biotic interactions also play a crucial role in shaping gorgonian distribution. Giant kelp (*Macrocystis pyrifera*) dominates many SCB reef systems, forming dense forests that structure the benthic environment by altering light availability, hydrodynamic flow, and competitive interactions (Schiel & Foster, 2015). While kelp forests provide habitat and food resources for numerous reef-associated organisms, their influence on sessile invertebrates such as gorgonians is not well understood. Some studies suggest that kelp compete for space and nutrients with sessile suspension feeders, while others indicate that kelp may facilitate certain benthic species by modifying local environmental conditions (Graham et al., 2007). In contrast, crustose coralline algae (CCA), a common encrusting group of calcifying algae, often co-occurs with gorgonians but may inhibit their settlement due to competition for space or through the production of allelopathic compounds (Harrington et al., 2004; Price et al., 2019). Investigating these interactions will provide valuable insight into the ecological dynamics of gorgonian populations in temperate reef systems.

b. Abiotic Stressors: Temperature and Productivity

In addition to biotic interactions, abiotic factors such as SST and primary productivity are likely to influence gorgonian density. Rising ocean temperatures, driven by climate change and periodic marine heatwaves, impose physiological stress on marine organisms, altering reproductive success, growth rates, and survival (Hughes et al., 2017). Gorgonians, which are adapted to cooler temperate conditions, may be particularly vulnerable to thermal stress, with prolonged exposure to elevated SSTs potentially reducing their recruitment and increasing mortality rates (Rossi et al., 2019). Meanwhile, primary productivity, indicated by chlorophyll-a concentration, may play an indirect role in shaping gorgonian distribution by influencing food availability and ecosystem productivity (Edmunds et al., 2020). Understanding how these abiotic factors interact with reef structural characteristics and biotic competition is essential for predicting gorgonian population dynamics under future climate scenarios.

c. Implications for Kelp Forest Ecosystems and Invasive Potential

Although gorgonians contribute to reef complexity, their increasing abundance in certain areas raises concerns about potential shifts in community structure, particularly within kelp forest ecosystems. While historically gorgonians have been a natural component of SCB reefs, changing environmental conditions and habitat modifications—such as the expansion of artificial reefs—may be facilitating their expansion into areas where they were previously less dominant. In regions where kelp populations have declined due to warming temperatures, sedimentation, or anthropogenic disturbances, gorgonians may proliferate, altering reef dynamics and potentially outcompeting kelp-associated taxa (Filbee-Dexter & Wernberg, 2018). Understanding where gorgonians are most likely to thrive is essential for predicting their future role in temperate reef ecosystems. If gorgonians expand unchecked into kelp-dominated habitats, they may reduce available settlement space for kelp holdfasts, alter hydrodynamic flow patterns, and shift trophic

interactions by providing habitat for different species assemblages. These changes could have cascading effects on species that rely on kelp forests for shelter and food, potentially reducing overall ecosystem resilience (Cavanaugh et al., 2019).

d. Study Objectives

This study aims to identify the key environmental and biological drivers influencing gorgonian distribution and density across natural and artificial reef habitats in the SCB. By integrating habitat characteristics, oceanographic conditions, and species interactions into a predictive framework, this research will provide a comprehensive assessment of gorgonian habitat suitability in temperate reef systems. Specifically, it will investigate the role of substrate heterogeneity, reef relief, SST, chlorophyll-a concentration, and competitive interactions with macroalgae in shaping gorgonian populations.

By addressing these knowledge gaps, this study will contribute to the broader understanding of temperate rocky reef ecology and inform ongoing efforts to manage and conserve these ecosystems. Given the increasing prevalence of artificial reefs as habitat restoration tools, understanding their ecological consequences for habitat-forming species like gorgonians is particularly relevant. Furthermore, by evaluating the potential for gorgonians to expand into degraded kelp forests, this research will provide insights into the resilience and long-term stability of temperate reef communities in the face of environmental change.

III. Materials and Methods

a. Study Sites and Data Collection

This study utilized data collected by the Vantuna Research Group (VRG) during annual SCUBA dive surveys conducted between 2016 and 2023. Surveys were performed at 77 sites of natural and artificial reefs throughout the SCB, spanning a range of environmental conditions and reef structural characteristics. These reefs represent diverse habitats, providing a comprehensive dataset for investigating the factors that influence the distribution and density of gorgonians. The study sites included reefs with varying degrees of environmental stress, substrate heterogeneity, and reef relief.

Additionally, this study incorporated ecological and oceanographic data for 49 sites from the Partnership for Interdisciplinary Studies of Coastal Oceans (PISCO), which collects biological, chemical, and physical data on ocean ecosystems in the nearshore portions of the California Current Large Marine Ecosystem. PISCO's data archive provides long-term, well-documented, and accessible ecological datasets, including depth-stratified sampling from the Northern Channel Islands (NCI). PISCO data for these sites were recorded numerically rather than using traditional depth zones, as the program does not use “deep” depth zones. These data were integrated into our final dataset to strengthen the analysis by broadening spatial coverage and increasing sample size.

MODIS-derived sea surface temperature (SST; °C) and chlorophyll-a concentration (CHL; mg m⁻³) data were obtained from the University of California San Diego, Scripps Institution of Oceanography Photobiology Group and collated. Monthly composite data were used instead of 15-day interval data employed in previous studies to provide a broader temporal resolution. Additionally, SST and CHL values were averaged over the period from 2016 to 2023 rather than calculating five-year averages preceding each study year.

b. Sampling Protocol

Dive surveys by the VRG followed the standardized Co-operative Research and Assessment of Nearshore Ecosystems (CRANE) protocol to assess gorgonian density across four depth zones: Inner Zone (~5 m), Middle Zone (~10 m), Outer Zone (~15 m), and Deep Zone (~20 m). This depth-stratified sampling method was applied consistently across all VRG study sites to ensure comprehensive coverage of varying reef habitats. However, the PISCO dataset does not categorize sites by depth zones but instead records depth numerically. In this study, we retained the depth zone format from the VRG dataset while analyzing the data using the *in situ* depth recorded by scientific divers. This allowed for the integration of both datasets, ensuring consistent depth information and facilitating a unified analysis of gorgonian habitat suitability across different reef types.

To quantify gorgonian density, the Uniform Point Contact (UPC) method was used in combination with band transects. Data were recorded at uniform intervals along each 30m transect to identify and enumerate key macroinvertebrate and macroalgae species, as well as structural complexity of the reef habitat (substrate heterogeneity and reef relief). Additionally, the percent cover of various substrate types (e.g., rock, sand, cobble) and biological communities (e.g., macroalgae, CCA) was recorded to assess benthic cover. Additionally, longitude and latitude coordinates for each dive site were recorded using GPS tracking on the dive boat to ensure spatial accuracy.

For giant kelp density, the CRANE approach used the SWATH (Swath Width and Transect Height) protocol, a method designed to estimate kelp density across diverse reef habitats. This approach involved swimming along designated 30m transects and using visual estimates to determine the density of kelp plants within specified width bands. The SWATH

sampling technique allowed for detailed spatial data collection, capturing variations in kelp distribution across the study sites.

c. Data Analysis

A linear mixed effects model (LMM) used to evaluate the relationship between gorgonian density and key environmental variables across 126 reef sites ($n = 3,850$ observations). The compiled dataset, including both VRG and PISCO data, was analyzed using R (version 4.3.2), a statistical software environment commonly used for ecological modeling and data visualization. The LMM incorporated both fixed and random effects, allowing for the inclusion of variability at site and regional levels while identifying the key factors influencing gorgonian density. Fixed effects included average depth (meters) as a numerical predictor for all sites, reef type (natural vs. artificial), percent cover of CCA, mean giant kelp density, substrate variability and standard deviation, and reef relief and standard deviation. Abiotic factors such as sea surface temperature (SST) and chlorophyll concentration (when available) were also considered as fixed effects. Random effects (Site and Region) accounted for variation across regions (e.g., Channel Islands, Santa Barbara Marine Zone) and site-specific differences that could not be explained by the fixed effects.

The model output was evaluated to determine the relative importance of each predictor variable in explaining gorgonian distribution patterns. Variance partitioning was used to assess the contributions of fixed and random effects to the overall variability in gorgonian density. To ensure that model assumptions were met, residual diagnostics were conducted. Model predictions were visualized to assess spatial patterns of gorgonian density across different regions of the SCB, specifically within natural and artificial reef habitats and across varying

depth zones. Cross-validation techniques were applied to evaluate the robustness and predictive accuracy of the model, ensuring that the results could be reliably interpreted. All analyses utilized the lme4 package for mixed-effects modeling, ggplot2 for data visualization, and additional packages such as caret and rsq for diagnostics and cross-validation, as well as data visualization software such as Biorender and Google My Maps for figure creation.

IV. Results

To determine the best-performing model, several alternative models were ranked using AICc values. The full model incorporating all predictor variables performed best (AICc = 32,703.6), while models with fewer variables, such as those testing only reef type (AICc = 44,057.3) or SST alone (AICc = 32,839.4), had substantially higher AICc values, indicating weaker explanatory power.

Several factors emerged as significant predictors of gorgonian density. Areas with greater substrate heterogeneity ($\beta = 12.51$, $t = 8.46$, $p < 0.001$) and higher mean reef relief ($\beta = 9.07$, $t = 5.86$, $p < 0.001$) supported higher gorgonian densities, reinforcing the idea that structural complexity enhances habitat suitability (Table 1, Figure 2). In contrast, more consolidated substrate types ($\beta = -8.80$, $t = -4.60$, $p < 0.001$) were associated with lower gorgonian densities, emphasizing the importance of habitat diversity (Table 1). Gorgonian density was significantly lower on natural reefs compared to artificial reefs ($\beta = -42.50$, $t = -1.96$, $p = 0.05$), suggesting that artificial reefs provide more favorable conditions for gorgonians, potentially due to increased structural complexity or recruitment surfaces (Figure 3). While SST showed a slight positive association with gorgonian density ($\beta = 2.09$, $t = 1.47$, $p = 0.14$), this effect was not statistically significant (Table 1). Increased CCA cover was associated with lower gorgonian

densities ($\beta = -0.06$, $t = -2.43$, $p = 0.015$), with interregional variability, suggesting competitive interactions between these organisms, particularly at greater depths (Figure 4).

Notably, giant kelp density did not have a significant effect on gorgonian density ($\beta = -3.71$, $t = -1.67$, $p = 0.10$), and there was no evidence of an interaction between kelp and depth. Similarly, chlorophyll-a concentration ($\beta = 0.22$, $t = 0.34$, $p = 0.73$) did not show a strong relationship with gorgonian density (Table 1).

V. Discussion

This study provides critical insights into the environmental and biological factors shaping gorgonian distribution and density across the SCB. By integrating biotic and abiotic variables into a predictive model, this research highlights key drivers of gorgonian habitat suitability and offers a foundation for future conservation efforts. Structural complexity, reef type, and competitive interactions with CCA emerged as the most influential predictors of gorgonian density, while sea surface temperature (SST) and kelp density showed weaker associations (Table 1). Our ranking of alternative models using AICc values confirmed that a model incorporating all predictor variables performed best, emphasizing the need to consider multiple interacting factors when assessing gorgonian habitat preferences.

a. Structural Complexity as a Key Determinant of Gorgonian Density

One of the most significant findings of this study is the strong positive relationship between gorgonian density and both substrate heterogeneity and mean reef relief (Figures 2A + 2B). These results reinforce the importance of structural complexity in creating suitable microhabitats for gorgonian recruitment and persistence. Increased substrate variability likely

provides a greater number of stable attachment points, reducing the risk of dislodgement and enhancing recruitment success. Similarly, reefs with higher relief may facilitate stronger water flow, which can improve suspension feeding efficiency in gorgonians by increasing access to particulate organic matter. These findings are consistent with previous research demonstrating that structurally complex habitats promote biodiversity and enhance the persistence of habitat-forming invertebrates (Linares et al., 2008; Perkol-Finkel et al., 2008).

b. Artificial Reefs as Potential Refugia for Gorgonians

Our results indicate that gorgonian densities were significantly higher on artificial reefs than on natural reefs, suggesting that artificial structures may serve as important refugia (Figure 3). The increased structural complexity of artificial reefs, combined with their typically lower sedimentation rates and higher relief, may provide more stable conditions for gorgonian recruitment and growth. These findings have important implications for habitat restoration and fisheries management, as artificial reefs are often deployed to enhance fish populations but may also provide secondary benefits for sessile invertebrates. Future studies should examine the long-term viability of artificial reefs as gorgonian habitats and assess whether these structures support similar levels of genetic connectivity and reproductive success as natural reef populations.

c. Thermal Stress and the Potential Impact of Rising SSTs

Although SST did not emerge as a statistically significant predictor of gorgonian density in this study ($p = 0.48$), the weak positive estimate suggests that temperature may still play a role under specific conditions. Previous research has indicated that temperature stress can negatively impact gorgonian growth, reproduction, and survival, particularly in warmer regions (Rossi et al., 2019). However, our findings do not provide strong evidence for a direct effect of SST on

gorgonian density in the SCB. Given that SSTs in the region are projected to rise due to climate change, future studies should investigate whether SST interacts with other environmental factors, such as substrate type, hydrodynamics, or nutrient availability, to indirectly influence gorgonian populations. Examining these interactions may offer a more nuanced understanding of gorgonian responses to climate variability and help predict potential long-term shifts in their distribution.

d. Biotic Interactions: The Roles of Giant Kelp and CCA

Contrary to expectations, giant kelp (*Macrocystis pyrifera*) density was not a significant predictor of gorgonian abundance, suggesting that direct competition between these organisms may be limited in this system. It is possible that other site-specific factors, such as wave exposure, nutrient availability, or localized hydrodynamics, mediate the effects of kelp on gorgonian populations. Further research is needed to determine whether these interactions vary across depth zones or in response to seasonal fluctuations in kelp biomass.

In contrast, our results indicate a significant negative relationship between CCA cover and gorgonian density (Figure 4). This suggests potential competitive interactions, where CCA-dominated substrates may be less conducive to gorgonian larval settlement, particularly with increasing depth. Given that CCA plays a crucial role in reef stabilization and coral recruitment, future studies should investigate whether the presence of CCA affects gorgonian settlement success or whether other environmental factors, such as wave action and sedimentation, drive this relationship.

e. Future Directions

Understanding the factors that influence gorgonian distribution is essential not only for conservation efforts but also for managing the balance between gorgonian populations and other key marine organisms, such as giant kelp, within the SCB. While this study emphasizes the role of substrate heterogeneity and reef relief in shaping gorgonian density, further research is needed to examine the complex interactions between gorgonians and giant kelp, particularly in terms of spatial competition and resource allocation. As slow-growing organisms that thrive in structurally complex environments, gorgonians may outcompete kelp in certain conditions, potentially leading to shifts in community structure. Investigating these interactions is crucial to prevent the overcolonization of kelp forests by gorgonians, which could disrupt these productive ecosystems.

Future studies should focus on how environmental factors, such as substrate type and reef complexity, affect gorgonian recruitment in kelp-dominated areas. Insights gained could inform management practices aimed at maintaining ecological balance, especially in regions where kelp forests are threatened by climate change, pollution, and other stressors. Additionally, research on the physiological limits of gorgonians in response to temperature fluctuations will be critical as rising sea surface temperatures (SSTs) may accelerate gorgonian growth, potentially expanding their range into kelp-dominated zones. Understanding the potential for such shifts will be vital for developing strategies that ensure the coexistence of both species without compromising the integrity of kelp ecosystems.

Moreover, exploring the genetic connectivity of gorgonian populations across natural and artificial reefs could provide insights into their potential for rapid dispersal, which may increase the risk of overcolonization in kelp forests. By identifying key dispersal mechanisms and connectivity patterns, researchers can better predict future gorgonian distribution and identify

regions at risk of overgrowth. Long-term monitoring programs that track gorgonian growth in relation to the health of kelp forests will be invaluable in understanding these dynamics and providing evidence-based recommendations for ecosystem management. This research is essential for developing targeted conservation strategies that safeguard the health and resilience of temperate reef ecosystems in the face of environmental change.

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Acknowledgements

I would like to express my deepest gratitude to the Undergraduate Research Center and the Virginia Reid Moore Foundation for their generous funding and support, which made this project possible. Special thanks to the incredible Vantuna Research Group (VRG) diving team—Chessie Craig, Julia McGlaughlin, Jess Peria, Daniel Ramirez, and Molly Brazinski—whose expertise and camaraderie helped me grow as a scientific diver. To my mentor, Dr. Dan Pondella, thank you for your unwavering support, insightful advice, and encouragement at every stage of this research. I also extend my sincere appreciation to Dr. Zoe

Kitchel and Chelsea Williams for their assistance with data analyses, which greatly strengthened this study. Finally, I am deeply thankful to my committee members, Dr. Shana Goffredi and Dr. Amanda Zellmer, for their thoughtful feedback and for helping me refine my scientific writing. This project would not have been possible without the collective effort of this incredible group of mentors and colleagues.

Appendix:

Table 1. General Linear Model Results for Predictors of Gorgonian Density

Results from a general linear model assessing the effects of environmental and habitat characteristics on gorgonian density (mean_density) in the SCB. Significant predictors ($p < 0.05$) are in bold.

Predictor	Estimate	Std. Error	t-value
Intercept	19.44	28.34	0.69
Mean Substrate SD	9.08	2.28	3.98
Mean SST (°C)	0.72	1.50	0.48
Giant Kelp Density	-3.71	2.22	-1.67

Percent CCA Cover	-0.06	0.03	-2.43
Mean Relief SD	4.46	2.87	1.55
Mean Relief	9.07	1.55	5.86
Mean Substrate	-8.80	1.91	-4.60
Mean Chl-a (mg/m ³)	0.22	0.64	0.34

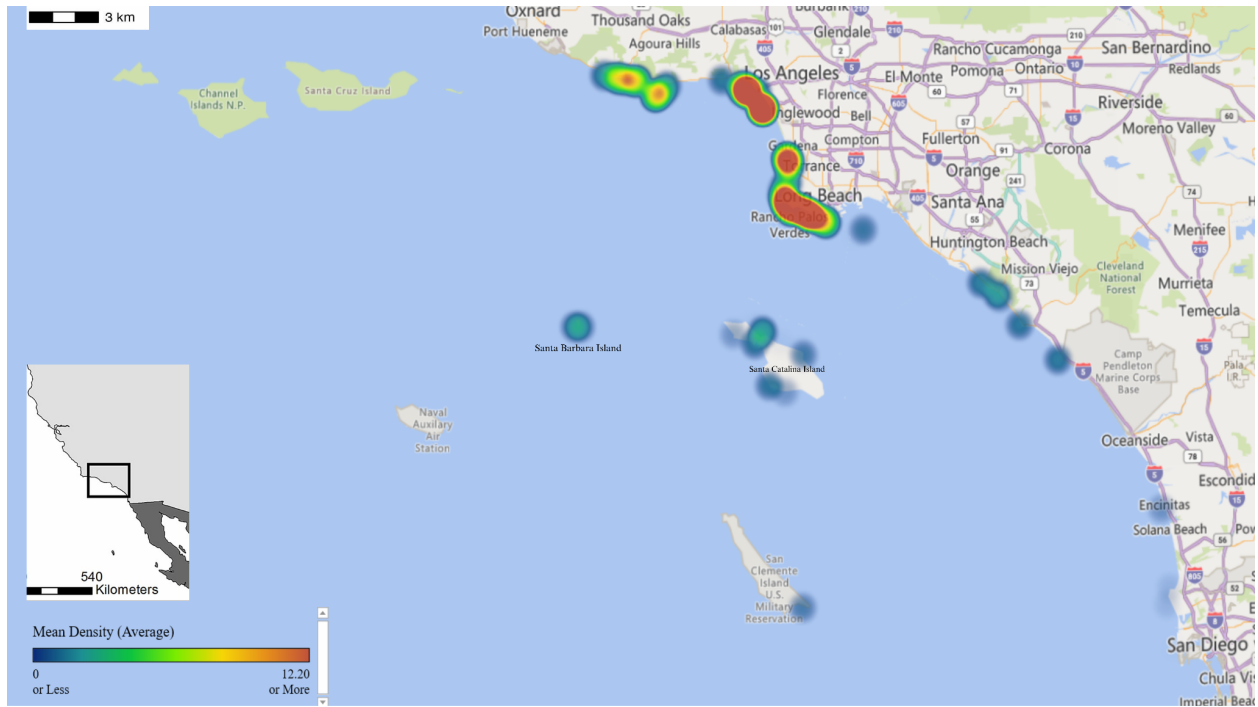


Figure 1. Heatmap of the SCB illustrating spatial variation in mean gorgonian density across survey sites, with color gradients ranging from blue (lowest density) to red (highest density). The map includes all Channel Islands. A reference image contextualizes the SCB's location within the United States. A scale bar in the top right corner provides distance context, and the latitude and longitude range extends from 32°43'N, 118°21'W (32.717°N, 118.350°W).

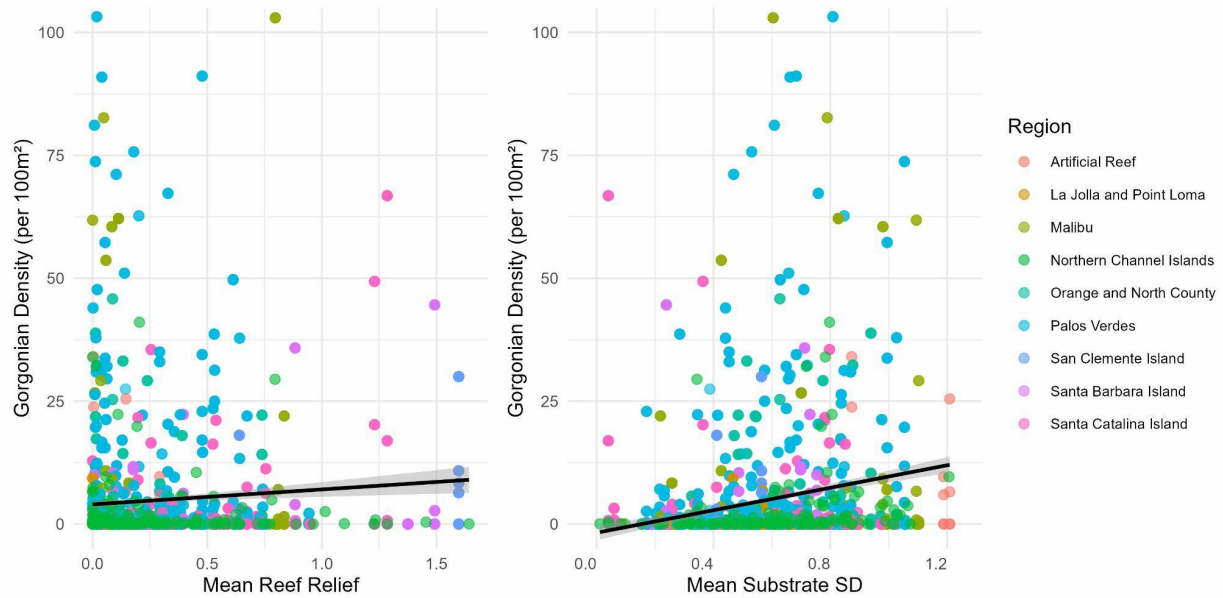


Figure 2. Partial regression plots showing the relationships between key habitat variables and gorgonian density across sites in the SCB. (A) Mean reef relief vs. gorgonian density, illustrating a positive association where increased reef complexity corresponds to higher gorgonian abundance. **(B)** Mean substrate standard deviation vs. gorgonian density, indicating how substrate variability influences gorgonian distribution. Points are colored by region, and regression lines (black) represent linear trends with confidence intervals.

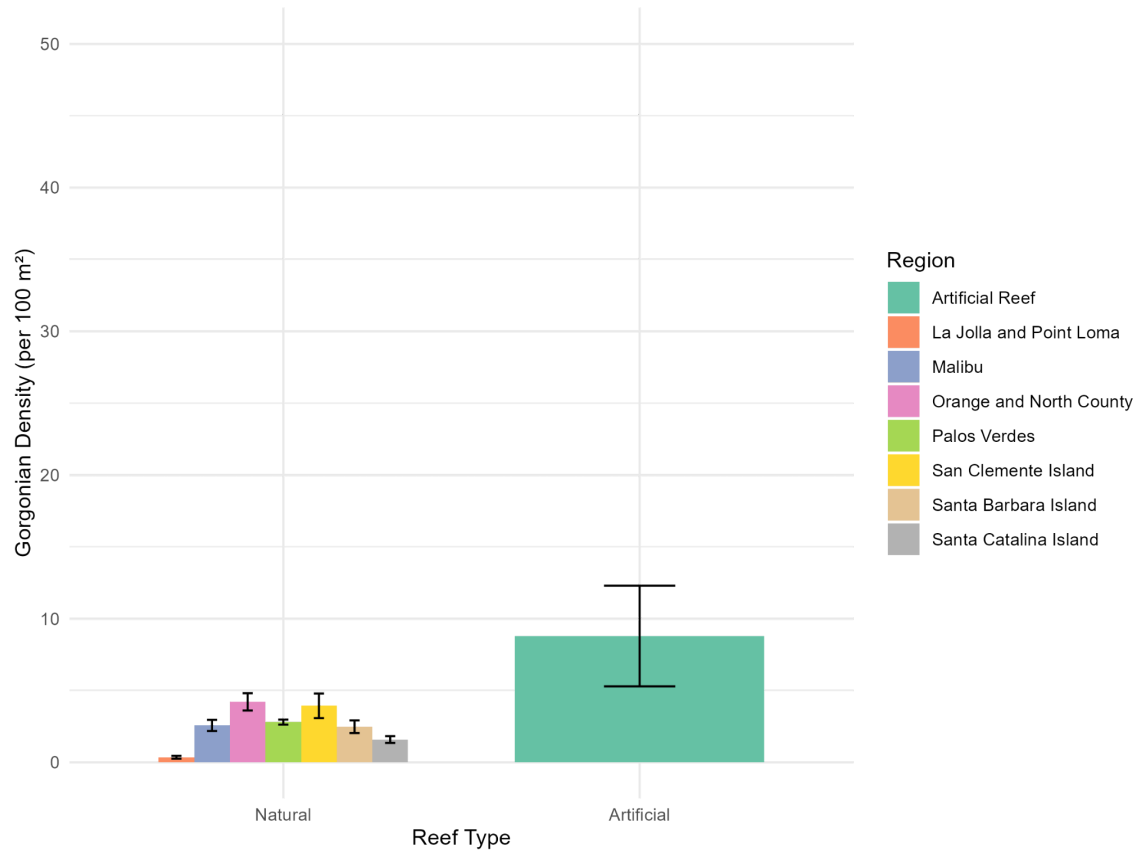


Figure 3. Gorgonian density (per 100 m²) across different reef types in the SCB, with data grouped by region. Bar graphs represent the mean gorgonian density, with error bars indicating the standard error of the mean. Data were separated by reef type (e.g., natural and artificial) and region, with the fill color representing regional differences. The y-axis is constrained to display values from 0 to 50 gorgonians per 100 m². Error bars were calculated from the standard error

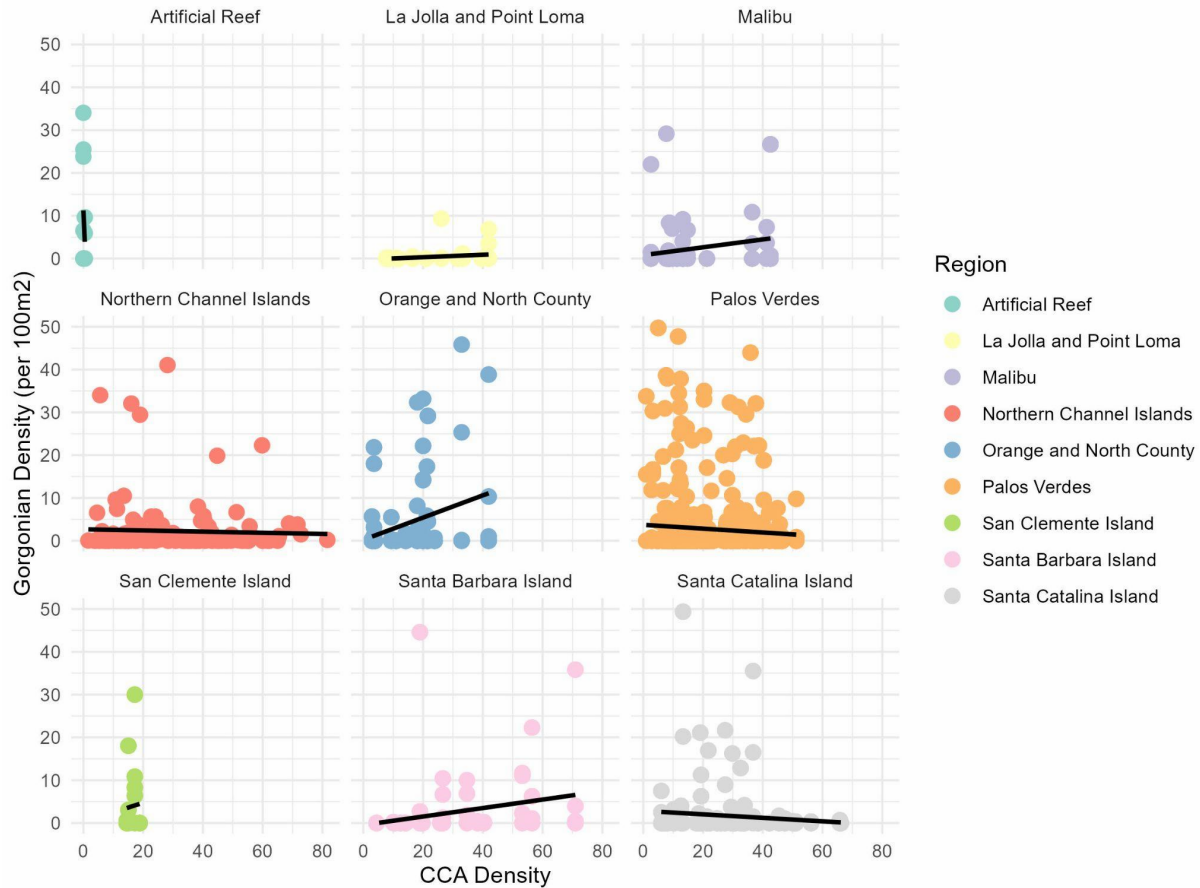


Figure 4. Scatterplot of gorgonian density (mean density, per 100 m²) versus CCA density (mean_brs, per 100 m²) across different regions. Each point represents the density measurements for a given site, with the points colored by **region**. A linear regression line (black) is fitted to the entire dataset, illustrating the relationship between gorgonian and CCA densities. Data are faceted by region, with each panel representing a distinct geographical region. The y-axis is constrained to values between 0 and 100 to focus on the range of observed gorgonian densities.